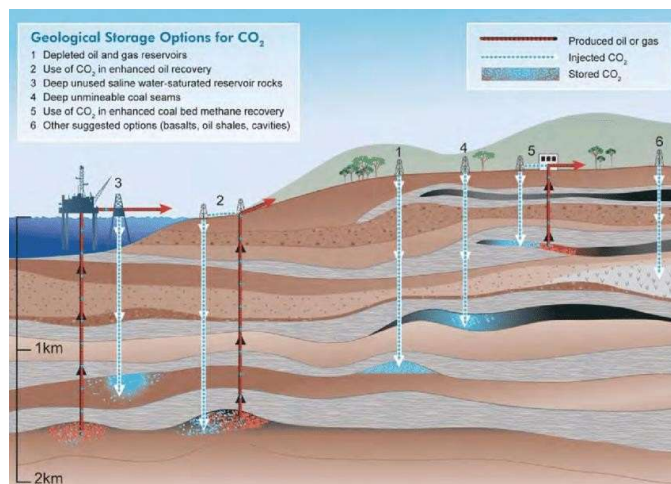


Geological Sequestration



Department of Chemical Engineering
National Institute of Technology, Warangal

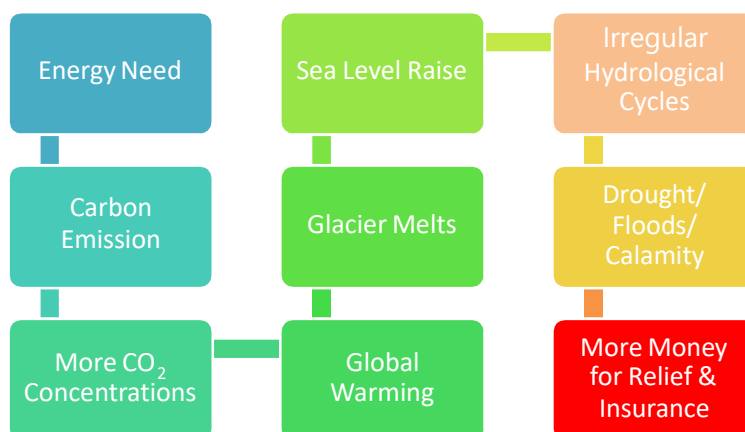
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1

Energy Problems – Financial Implications

- The economy is projected with an average of 10% per annum in India. This increases the carbon emissions exponentially.



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Carbon Capture Options



- Capturing CO₂ from the major stationary sources and its storage into deep geological formations is considered as a potential mitigation option.

Depleted Oil and Gas Reservoirs

Enhanced Oil Recovery

Enhanced Coal Bed Methane Recovery

Deep Unmineable Coal Seams

Unused saline water-saturated formations

Geological sites – shales, basalt rocks, cavities

Oceans

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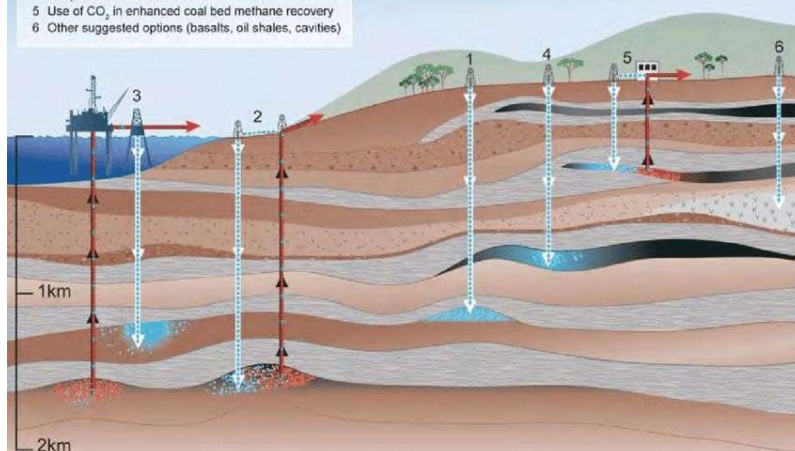
Carbon Capture Options



Geological Storage Options for CO₂

- 1 Depleted oil and gas reservoirs
- 2 Use of CO₂ in enhanced oil recovery
- 3 Deep unused saline water-saturated reservoir rocks
- 4 Deep unmineable coal seams
- 5 Use of CO₂ in enhanced coal bed methane recovery
- 6 Other suggested options (basalts, oil shales, cavities)

Produced oil or gas
Injected CO₂
Stored CO₂



Peter et al. (2009). Opportunities for underground geological storage of CO₂ in New Zealand -Report

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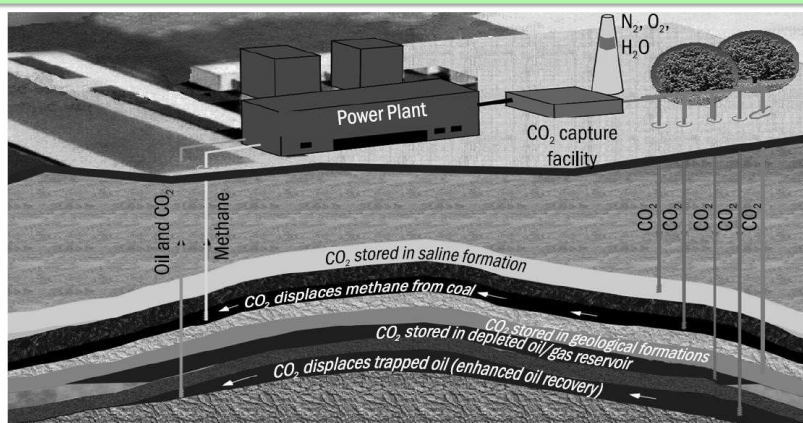
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Carbon Capture Options



Enhanced Oil Recovery



Viable options for CO₂ storage

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Carbon Capture Options



Enhanced Oil Recovery (CO₂ - EOR)

- **Miscible Phase** – CO₂ mixes with oil and reduces the viscosity of oil to flow through the well.
- **Immiscible phase** – CO₂ gas is used to re-pressure the reservoir as a sweep gas to move the oil towards the production well.
- **Weyburn-Midale project** (Canada) : Injects daily 5000 tonnes of CO₂/day and increased the oil productions by 10000 barrels/day.
- At the end of the project 30 Million Tonnes of CO₂ will be stored permanently and producing at least 130 million barrels of enhanced oil recovery.
- India – Oil and Natural Gas Corporation Ltd. (**ONGC**) proposed to use the same in Ankleshwar field.
- The proposed potential is storage of 5 – 10 Million Tonnes of CO₂ along with enhanced oil recovery.

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Carbon Capture Options



Enhanced Coal Bed Methane Recovery

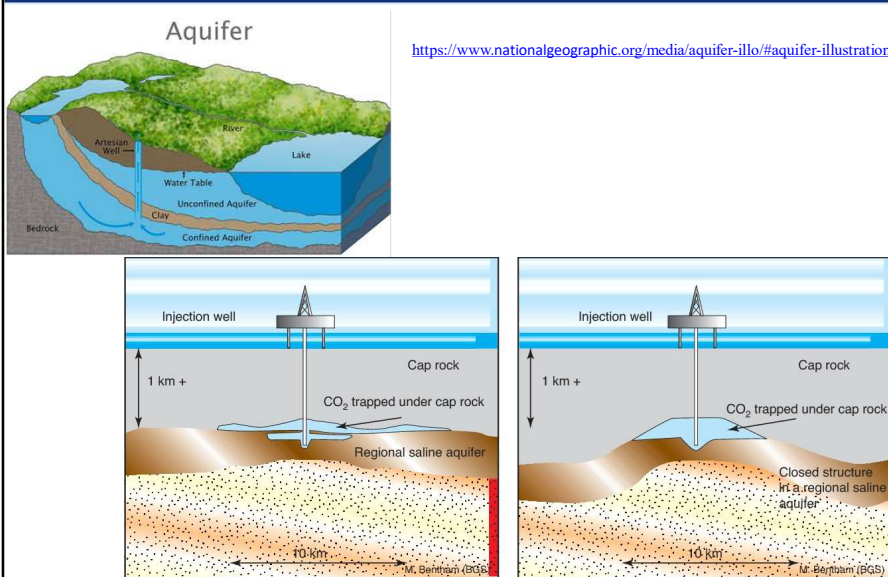
- Coal beds typically contain large amounts of methane absorbed onto the surface of the coal.
- CO₂ gas efficiently replaces the CH₄ molecules with a selectivity of **2:1** from the internal surface of coal layers.
- India having lot of coal beds are having lot of opportunities with some limitations.
- **Coal rank** – The adsorption of CO₂ is a function of the amount and reactivity of surface area in pores which is a function of coal rank.
- The coal should have minimum amount of **minerals** like vitrinite type.
- Coal of different **geological ages** have different compaction which vary in porosity.
- The higher depth and thickness of coal seams provides more volume for storage.

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Carbon Capture – Saline Aquifers



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Carbon Capture – Saline Aquifers



- Estimated storage capacity of saline formations is large enough – **viable long term storage solutions**.
- Many parts of the world has saline aquifers – reduces the **cost of infrastructure** at stationary polluting sites.
- They can help in reaching **near zero emissions** for existing power plants.
- Natural storage options – more **credible storage**.
- The sequestered carbon is impervious through the solid cap rock so **more residence time** in the bed.
- Limited possibility of damages and negative impacts.
- The cost is limited because of higher interaction of CO₂ with the saline formations.
- The most suitable reservoirs are those at **depths greater than 800 m**, as CO₂ will behave more like a liquid than a gas, enabling much more quantity to be stored.

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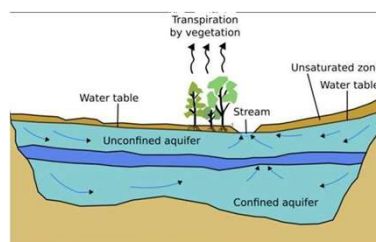
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Carbon Capture – Saline Aquifers



- For storage site – **depth of the aquifer** and **cap rock** are critical factors.
- The aquifers are selected with minimum of 800 to 1000 m below surface level where CO₂ exists in liquid phase.

1000 Kg CO ₂ Gas	0 m	0 °C & 1 bar	509 m³
	1000 m	35 °C & 102 bar	1.39 m³



- The reservoir rock must be covered by a **cap rock** which is impermeable to CO₂. Cap rocks are usually **shales, mudstones** and must be non-faulty.
- The mechanisms involve physical trapping, dissolution and mineralization.
- Unconfined aquifers – Physical or geochemical trapping
- Confined aquifers – stratigraphic and structural trapping

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Stratigraphic and Structural Trapping

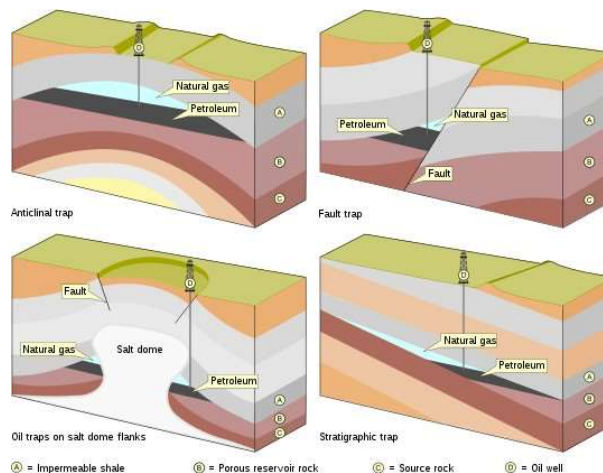


➤ Structural traps :

Hydrocarbon traps that form in geologic structures such as folds and faults

➤ Stratigraphic traps:

Hydrocarbon traps that result from changes in rock type or pinch-outs, unconformities, or other sedimentary features such as reefs or buildups.



https://commons.wikimedia.org/wiki/File:Oil_traps.svg

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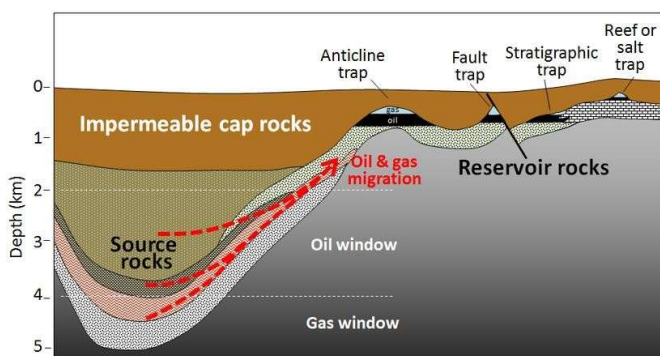
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Physical Trapping : Hydrodynamic



- This type involves the storage of CO₂ in unconfined aquifers. Once gas is injected it moves along the way till it touches the cap rock.
- After that the gas moves laterally filling all the domes and crusts wherever possible.
- This creates CO₂ distribution over a large area under cap-rock reservoir boundary with low concentrations.



https://energyeducation.ca/encyclopedia/Source_rock

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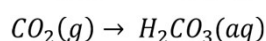
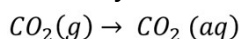
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Geochemical Trapping

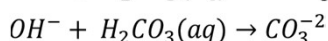
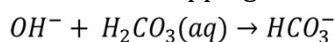


- CO_2 undergoes a series of chemical reactions under the sub surfaces with water, minerals etc.

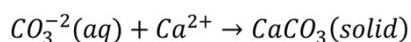
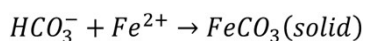
Solubility reactions



Ionic trapping



Mineral Trapping



- Mineral trapping is relatively slow process ~ thousands of years or long

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Saline Aquifers Storage Issues



- **Reservoir properties modelling** – requires extensive study for characterization and modelling of reservoirs. Test models and migration paths will give data but with **expenditure**.
- **Cap rock integrity** - determination of cap rock properties by testing and modelling is likely to be required.
- **Aquifer flow modelling** – Aquifers flow from high pressure to low pressure areas. **Flow modelling** and CO_2 leakage should be considered.
- **Solubility in groundwater** – 2 mol% in solution. The solubility decreases with increase in temperature.
- **Reaction with host track** – minerals dissolved or higher pressure cases stability in terms of fracturing. This will increase porosity.
- **Ground water pollution** - Contamination might happen if CO_2 is escaped because of a seal failure or leak in injection system or faults existed.

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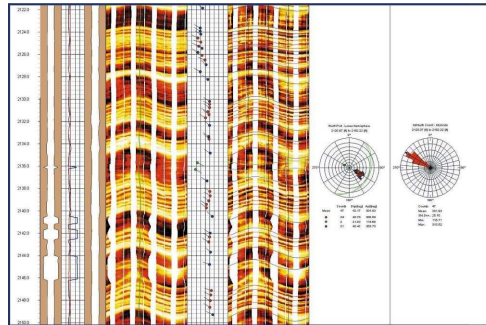
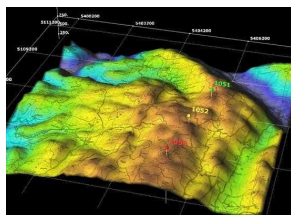
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Saline Aquifers Screening



- All sedimentary basins might not be useful for storage because some are shallow and some are filled with more rocks with less permeability.
- In general, storage sites should have adequate **Capacity and injectivity**, Satisfactory **sealing cap** rock and confirming unit, and **Stable geological environment**.
- **Sub-surface geological** structures
 - ✓ Bore hole analysis
 - ✓ Geophysical surveys



<https://geotechpedia.com/Software/Show/623/BOREHOLE-IMAGE-ANALYSIS#prettyPhoto>

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Saline Aquifers Screening

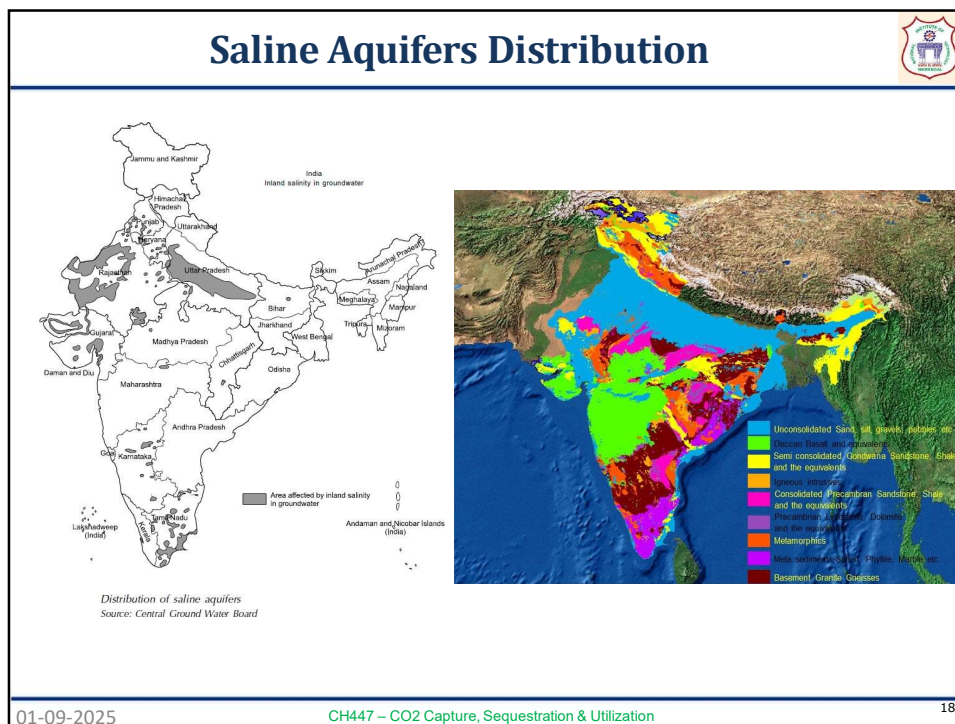
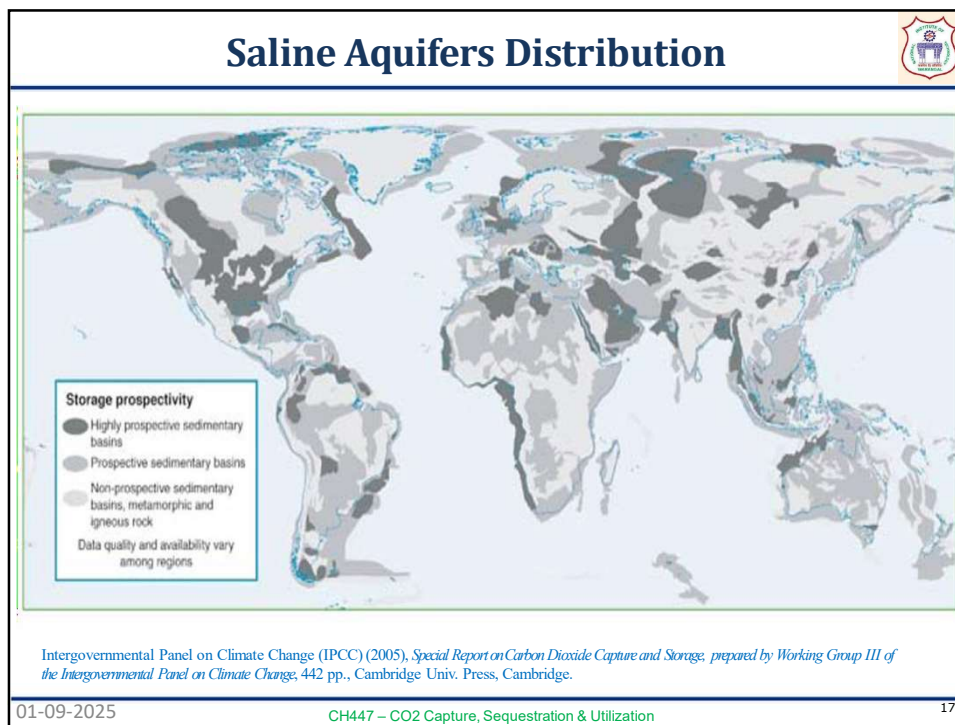


- **Screening criteria:**
 1. Safe storage of CO₂ depends on **location assessment**, target reservoir characteristics, stratigraphy sequence, deposition model.
 2. CO₂ trapped is assumed to remain in **trapped state for thousands of years** without any leakage.
 3. The pore spaces must be large enough to give **bulk volumes of CO₂ storage** with in a site.
 4. Barriers in fluid flow and **impermeable rocks** make the site less suitable for CO₂ storage. They limit the amount of CO₂ that can be pumped inside the site location.
 5. The target site should be below 800 m from surface so that the **CO₂ is in bulky liquid phase** which occupies a less volume.
 6. The suitable location should be free form **tectonic movements** or earth quake zone.
 7. The **separation** for water and CO₂ should be adequate in given location.

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Saline Aquifers Distribution



Global storage capacity for geological storage options

Geological storage option	Global capacity	
	Reservoir type lower estimate of storage capacity (Gt CO ₂)	Upper estimate of storage capacity (Gt CO ₂)
Depleted oil and gas fields	675	900
Unminable coal seams	3–15	200
Deep saline reservoirs	1000	Uncertain, but possibly 10 ⁴

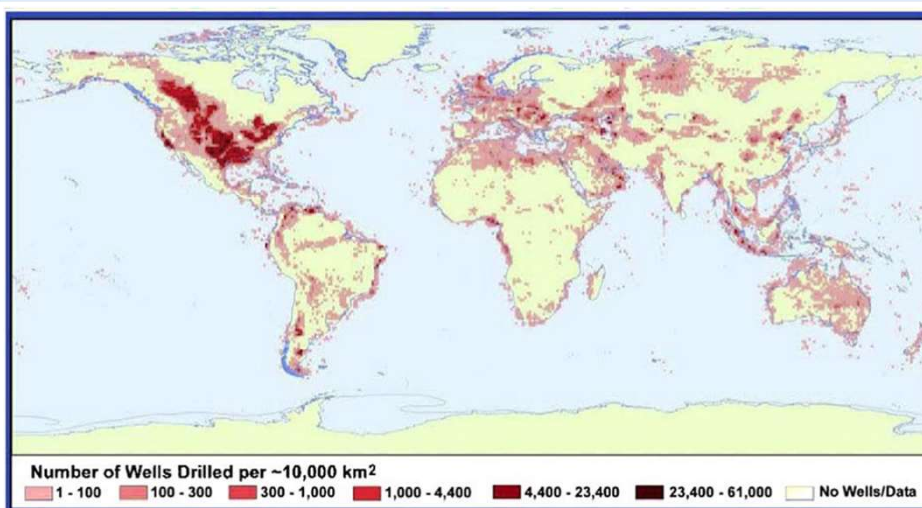
- Almost two-third parts of the country occupied by the Archean igneous and metamorphic rocks have negligible porosity.
- India has saline aquifers ranging from 30 m to 450 m below surface and having several lakhs of km² area distribution.

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Saline Aquifers Drills

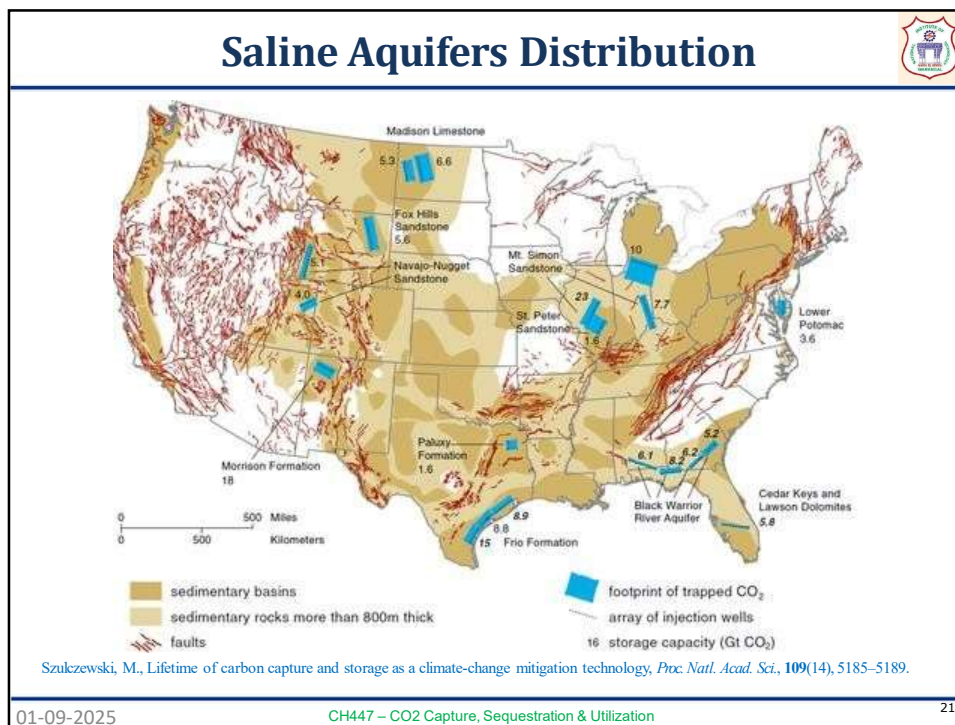


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Carbon Capture Options



Basalt Formations

- Basalt rocks – promising irreversible long term storage of CO₂.
- CO₂ is injected in **supercritical state** (liquid and gas properties combination) – adsorbed **hydrodynamic, solubility, and mineral trappings**.
- Hydrodynamic trapping – lower storage integrity, solubility trapping – better efficiency and Mineral trapping is most effective method.
- Basalt rocks have solid cap rocks and sites for geochemical reactions to take place with **high level of storage security**.
- Laboratory study – *in-situ* mineralization reaction is less than 1000 days.
- Study of PNNL USA and NGRI India - Deccan traps are most suitable for storage in this method.
- Deccan traps are situated on seismic zones of type II and III.

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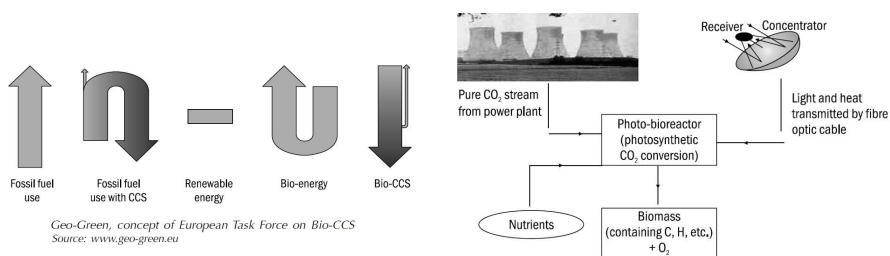
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Carbon Capture Options



Bio-Carbon Capture and Sequestration (Bio-CCS)

- A closed loop in which CO₂ is removed from atmosphere by plants which are converted into biofuels as energy source.
- Use of Bio-Carbon capture predicted negative emissions by 2050.
- **Bio-sequestration** – converts CO₂ from power plants into biomass and oxygen in a photo bio-reactor.



Geo-Green, concept of European Task Force on Bio-CCS
Source: www.geo-green.eu

Photosynthetic conversion of CO₂ into biomass
Source: Energy Conversion and Management Vol. 46, February 2005, 403–420

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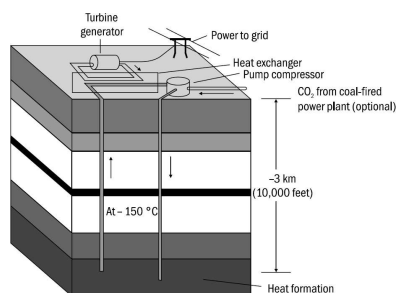
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Carbon Capture Options



Other options form CO₂

- CO₂ can be utilized as heat exchanger fluid as it extracts more heat than water in geo-thermal system.



Geothermal power using CO₂ as the heater fluid
Source: www.ecogeek.org

- Sea Urchins use nickel ions to capture CO₂ to develop the exoskeleton – solid carbonate minerals.



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